Ratul Hasan

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EDUCATION

Bangladesh Army University of Engineering & Technology (BAUET) *Natore, Bangladesh*
B.Sc. in Computer Science and Engineering | CGPA: 3.82/4.0 *2022 – Jul 2026*

Thesis Title: Biology-Informed Recurrent Neural Networks for Longitudinal Hippocampal Atrophy Forecasting in Alzheimer's Disease [J1]

- Designed a biology-informed loss constraining LSTM/GRU forecasts to monotonic decline, noise-bounded growth and plausible atrophy rates, using FastSurfer hippocampal volumes of 382 ADNI subjects over 36 months
- Bio-Informed GRU outperformed ARIMA, XGBoost and unconstrained RNN baselines (MAE 115.0 mm^3, $r = 0.994$) with 0% biologically implausible predictions on the held-out test set, vs. 3.9–13.0% for baselines

Govt. Majid Memorial City College *Khulna, Bangladesh*
Higher Secondary School Certificate | GPA: 5.0/5.0 *2019 – 2021*

IELTS Academic: Overall 7.0 (Listening 7.5, Reading 9.0, Writing 6.0, Speaking 6.0)

PUBLICATIONS

Conference Papers

[C1] A Gas-Optimized Blockchain Framework for Scalable Document Verification Using Dynamic Access Control and IPFS *2026*
R. Hasan, S. Farjana, Y. Oley, M. O. Pranto and M. A. Rayhan. *IEEE PECCII 2026.*
doi:10.1109/PECCII70991.2026.11662071 | Presentation Certificate

[C2] SplitSafe Ultra: A Moving-Target Defense Framework for Entropy-Aware Encrypted File Persistence on IPFS *Presented*
M. A. Muhaimin, **R. Hasan**, Y. Oley, S. Farjana, B. R. Roy, S. K. Paul and M. E. Hamid. Accepted & presented at *IEEE SPICSCON 2026*; to appear in proceedings.
Presentation Certificate

[C3] ActiVisionNet-Fusion: A Dual-Stream Explainable Deep Learning Framework for Driver Drowsiness Detection *Presented*
N. Ahmed, **R. Hasan**, S. K. Paul, D. N. I. Noor, M. M. Haque and M. E. Hamid. Accepted & presented at *IEEE SPICSCON 2026*; to appear in proceedings.
Presentation Certificate

[C4] Quantum-Resistant FOTA: End-to-End Decentralized Firmware Updates for IoT Using Blockchain and CRYSTALS-Dilithium *2025*
R. Hasan, M. M. Haque, R. Haque, Y. Oley and R. Akter. *ACM NSysS '25*, pp. 110–114.
doi:10.1145/3777555.3777569

Journal Manuscripts

[J1] BI-RNN: Biology-Informed Recurrent Neural Networks for Longitudinal Hippocampal Atrophy Forecasting in Alzheimer's Disease *Under Review*
R. Hasan, M. M. Haque, A. Sarker and Y. Oley. *Scientific Reports*, submitted May 2026. Based on B.Sc. thesis.

[J2] Entropy-Assured Post-Quantum Key Generation on IoT Edge Devices: Source Independence, Measured Min-Entropy and Signature Scheme Selection *Under Review*
R. Hasan, M. A. Rayhan, M. O. Pranto, M. M. Haque, Y. Oley and S. Farjana. *PLOS ONE*, submitted Sep 2026.

[J3] Methodological Rigour and Risk Evaluation of Large Language Models in Mental Health: A Systematic Review *Under Review*
R. Hasan, M. Hoque, M. Hossain, M. K. Morol and T. H. Liew. *Frontiers in Digital Health*, submitted Nov 2026.

Preprints

[P1] Disentangling Label-Space Shift from Domain Shift in Federated Continual Learning for Mental Health Text Classification *2026*
R. Hasan and M. Hoque. *SSRN*, 2026.
doi:10.2139/ssrn.7662638

EXPERIENCE

ELITE Research Lab LLC | *Remote* — *Queens, NY, USA*

- ***Research Assistant*** — *Oct 2026 – Present*
  Supervisor: Md Kishor Morol | Recognition Certificate
  - Contributing to a journal study on prompt-injection risks in LLM smart-home agents
  - Running experiments on how such attacks alter agents in a simulated home
  - Evaluating defences and their trade-off between security and task performance
- ***Research Intern*** — *Jun 2026 – Sep 2026*
  Supervisor: Md Kishor Morol | Completion Certificate
  - First-authored a systematic review of 82 studies on LLMs in mental health, appraising methodological rigour and risk reporting (hallucination, bias, safety, human evaluation) with a 24-field extraction instrument
  - Studied catastrophic forgetting in federated continual learning for mental health text classification, separating label-space shift from domain shift on Reddit data
  - Benchmarked seven training regimes, including FedAvg, FedProx, EWC, replay and centralized baselines, across 10 clients and three seeds with a MiniLM encoder

BAUET Computer Society, Dept. of CSE | ***Programming Instructor*** — *Natore, Bangladesh*
Experience Letter — *Jul 2024 – Dec 2025*

- Volunteer-taught C, C++ and data structures to undergraduates over three semesters
- Ran coding exercises and programming contest practice sessions

Lab AR | ***Software Engineering Intern*** — *Dhaka, Bangladesh*, *Jul 2025 – Aug 2025*

- Led a team project to develop a web-based Property Management System
- Built backend features in Laravel (PHP)

PROJECTS

LedgerSeal: Blockchain-Based Document Verification on IPFS — *GitHub*

- Built a DApp that stores documents on IPFS and verifies them via constant-time on-chain hash lookups, independent of dataset size, with role-based upload control on an EVM-compatible blockchain
- Cut amortized storage cost 21.8% via batch uploads (41,317 → 32,297 gas/document) and duplicate-hash reverts; validated with 1,000 fuzz runs and a 10,000-document load test [C1]. *Tools: Solidity, Foundry, React, IPFS*

BAUET Project & Thesis Archive — *GitHub*

- Built a Django (MVT & REST) archive indexing BAUET theses and projects. *Tools: Django, PostgreSQL*

HONORS & COMPETITIVE PROGRAMMING

Online Judges

- Codeforces: Expert, max 1650 (**Ratul_Hasan**) | CodeChef: 3 Star, max 1676 (**ratulhasan2108**)

Programming Contests

- ICPC Asia Dhaka Regional Contestant (Team: BAUET_Rong_Answer)
- National Collegiate Programming Contest (JU): Rank 149 (Team: BAUET_PARADIGM)
- Inter-University Programming Contest (twice): Best Rank 45 at IUT (Team: BAUET_Twisted_Minds)
- Ranked 10th in Independence Day Programming Contest by Military Institute of Science & Technology
- Intra-University Programming Contest: Champion (Team), Runner-up (Solo)
- MIND STORM 4.0 Programming Contest, BAUET: 2nd Runner-up

Others

- 1st Runner-up, Intra-University Three Minute Thesis (3MT) Presentation

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, SQL, Solidity
ML & Research: PyTorch, Hugging Face (Transformers)
Web & Blockchain: Django, Django REST Framework, Laravel, React, Foundry, IPFS
Tools & Databases: Git, GitHub, Linux, PostgreSQL, MySQL
Research Interests: AI in Healthcare, Cybersecurity, Cloud Infrastructure

VOLUNTEERING & LEADERSHIP EXPERIENCE

- President, BAUET Computer Society (promoted through five executive roles)
- Problem Setter, Intra-University Programming Contest, BAUET
- Officer of Documentation, Hult Prize at BAUET